

A Reduced-Precision Network for Image Reconstruction

MANU MATHEW THOMAS, University of California, Santa Cruz

KARTHIK VAIDYANATHAN, Intel Corporation

GABOR LIKTOR, Intel Corporation

ANGUS G. FORBES, University of California, Santa Cruz



Fig. 1. A comparison of different reconstruction techniques on a frame from the INFILTRATOR scene with PSNR (top row) and SSIM (bottom row) metrics. TAA has the worst reconstruction quality with excessive blurring followed by direct prediction with U-Net. QW-Net produces the best results with quality comparable to brute force sampling with 256 samples per pixel.

Neural networks are often quantized to use reduced-precision arithmetic, as it greatly improves their storage and computational costs. This approach is commonly used in image classification and natural language processing applications. However, using a quantized network for the reconstruction of HDR images can lead to a significant loss in image quality. In this paper, we introduce *QW-Net*, a neural network for image reconstruction, in which close to 95% of the computations can be implemented with 4-bit integers. This is achieved using a combination of two U-shaped networks that are specialized for different tasks, a *feature extraction* network based on the U-Net architecture, coupled to a *filtering* network that reconstructs the output image. The feature extraction network has more computational complexity but is more resilient to quantization errors. The filtering network, on the other hand, has significantly fewer computations but requires higher precision. Our network recurrently warps and accumulates previous frames using motion vectors, producing temporally stable results with significantly better quality than TAA, a widely used technique in current games.

CCS Concepts: • **Computing methodologies** → **Neural networks; Rendering.**

Additional Key Words and Phrases: Kernel prediction, antialiasing

Authors' addresses: Manu Mathew Thomas, University of California, Santa Cruz, mthomas6@ucsc.edu; Karthik Vaidyanathan, Intel Corporation, karthik.vaidyanathan@intel.com; Gabor Liktó, Intel Corporation, gabor.likto@intel.com; Angus G. Forbes, University of California, Santa Cruz, angus@ucsc.edu.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

© 2020 Copyright held by the owner/author(s).

0730-0301/2020/12-ART231

<https://doi.org/10.1145/3414685.3417786>

ACM Reference Format:

Manu Mathew Thomas, Karthik Vaidyanathan, Gabor Liktó, and Angus G. Forbes. 2020. A Reduced-Precision Network for Image Reconstruction. *ACM Trans. Graph.* 39, 6, Article 231 (December 2020), 12 pages. <https://doi.org/10.1145/3414685.3417786>

1 INTRODUCTION

Synthesizing computer-generated imagery involves two key functions: sampling the incident radiance on an image plane and applying a reconstruction filter to produce the final image [Cook et al. 1984]. If the incident radiance is undersampled, high-frequency components resulting from visibility discontinuities or specular lighting can appear as aliasing in the reconstructed image. To reduce aliasing, production renderers typically apply supersampling with a large number of samples per pixel. On the other hand, real-time applications rely on a combination of radiance prefiltering and specialized image reconstruction techniques. Temporal antialiasing (TAA) [Jimenez et al. 2012; Karis 2014; Yang et al. 2009] is an image reconstruction technique that is widely used in games today. TAA uses renderer-generated motion vectors to gather and accumulate samples from previous frames, effectively increasing the number of samples per pixel. However, TAA is susceptible to ghosting artifacts, loss of detail, and temporal instability [Yang et al. 2020].

Convolutional neural networks offer a promising alternative for image reconstruction, and have been successfully applied to closely related fields such as denoising [Bako et al. 2017; Chaitanya et al. 2017; Vogels et al. 2018]. The U-Net [Ronneberger et al. 2015] architecture is particularly well suited for such tasks as it processes features at multiple scales, achieving a large receptive field and more effective filtering with a relatively low computational cost.